Lumpy Adjustment: New Evidence and Aggregate Implications

Work in Progress

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Intro

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- Firms do not invest a little bit every year (not continuous).
- Instead, they invest in *big, infrequent jumps* when they expand, modernize, or upgrade.
- They often cluster across firms and can move the whole economy.

  We study whether, and how, these investment spikes combine to drive business cycles.
- Build a twow-way empirical toolkit to answer this questions micro $\leftrightarrow$ macro variables

Key Terms

- **Extensive margin** $\rightarrow$ How many firms decide to invest this period.
- **Intensive margin** $\rightarrow$ How much each investing firm spends when it invests.
- **Impulse response** $\rightarrow$ How firms or the economy adjust over time after a shock hits.

Overview

- Firms adjust capital in *large, lumpy jumps* — not smooth increments.
- These jumps tend to **cluster**: many firms invest around the same time.
- When this clustering happens, **aggregate business cycle movements emerge**.
- We separate investment into "lumpy" vs. "non-lumpy" components.
- Only the **lumpy component** reacts strongly to economic shocks.
- Therefore, the **extensive margin** — how many firms invest — is the key micro foundation of the business cycle.

Lumpiness at the Firm Level - Compustat U.S. Data 1980-2022

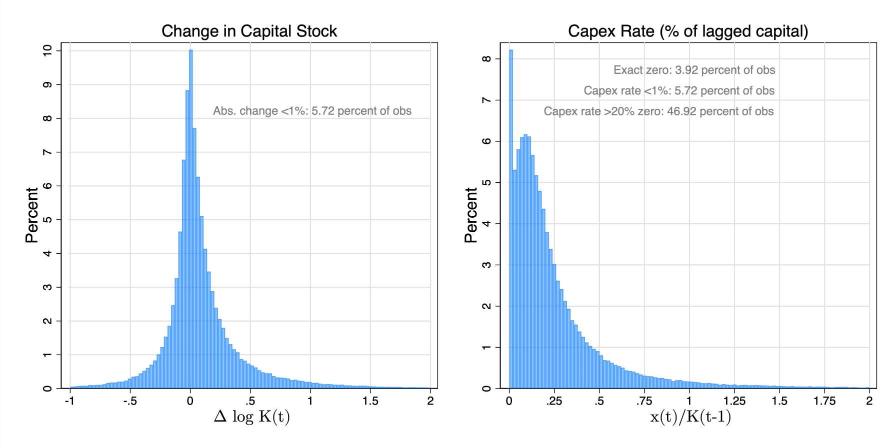


Figure 1: Lumpiness in U.S. Compustat Data of Public Firms

Variables

Main Firm-Level Variables:

- Standard measurement following the literature (Ottonello and Winberry (2020)).
- **Capital** ($\boldsymbol{K_{it}}$): Measured as PPE under the perpetual inventory method.
- **Investment** ($\boldsymbol{x_{it}}$): Capital expenditure (capex) spending.
- **Firm Investment Rate** ($\frac{\boldsymbol{x_{it}}}{\boldsymbol{K_{it-1}}}$): The ratio of current capex to lagged capital.
- **Aggregated Measures:**

$$I_{\text{agg}} = \sum_i x_{it}, \tag{1}$$

$$K_{\text{agg}} = \sum_i K_{it-1}. \tag{2}$$

Lumpy Variables

Aggregated Lumpy Variables:

- **Lumpy Investment**: The aggregate capex of firms that make lumpy adjustments,

$$I_{20} = \sum_{i \in \mathcal{L}} x_{it}, \tag{3}$$

where $\mathcal{L}$ denotes the set of firms that make lumpy capital adjustments.

- **Lumpy Adjusters' Capital Stock**: The total lagged capital for these firms,

$$K_{20} = \sum_{i \in \mathcal{L}} K_{it-1}. \tag{4}$$

Extensive, Intensive margins of Lumpy Adjustment

- Following Guorio and Kashyap (2007), lumpy events can be decomposed into extensive and intensive margins

$$
\begin{aligned}
log\frac{I20}{Kagg} &= log\frac{I20}{K20} + log\frac{K20}{Kagg} && (5)\\
&= log\text{-}IPA + log\text{-}ADJ && (6)\\
&= size\ of\ adjustment + share\ adjusting && (7)
\end{aligned}
$$

- IPA - investment per adjuster (Intensive Margin)
- ADJ - number of firms adjusting (Extensive Margin).

Agg Investment Rate = Lumpy + Nonlumpy Aggs

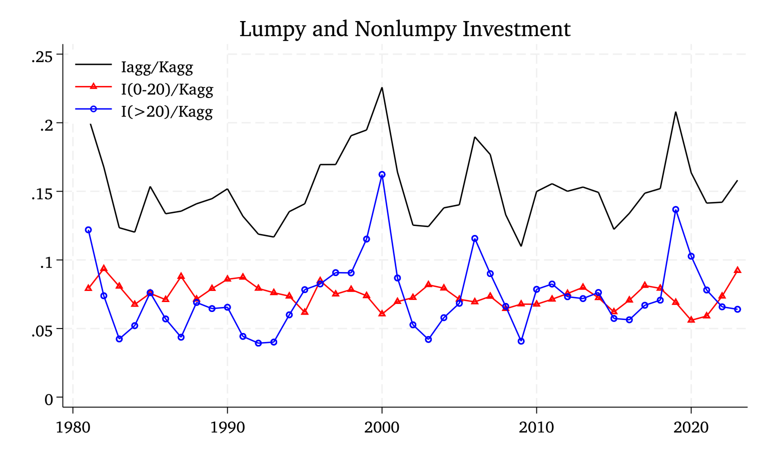


Figure 2: Investment Rate (Iagg/Kagg), Non Lumpy Investment Rate (I(0 − 20)/Kagg) and Lumpy Investment Rate (I(> 20)/Kagg

- Agg investment rate and its "Lumpy" part seem to co-move.

Lumpy Adjustments = Intensive + Extensive Margin

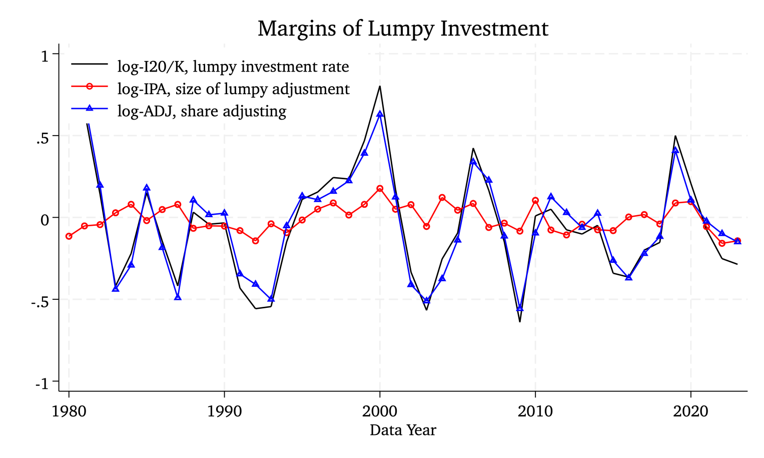


Figure 3: Lumpy Adjustments (I20/K) = Intensive Adjustments (IPA) + Extensive Adjustments (ADJ)

- Extensive Margin - the share of firms adjusting lumpily seem to be more important than the Intensive Margin - the size of lumpy

Aggregate Implications

- Aggregate investment is not lumpy.
- The cross-sectional distribution of capital holdings is potentially important.
- **Does it matter for the aggregate?**
    - Thomas (JPE, 2002) and Khan and Thomas (Econometrica, 2008) show that distribution of capital is irrelevant for aggregate investment and output. GE effects: consumption smoothing affects interest rates and desired capital!
    - Bachmann, Caballero and Engel (AEJ: Macro, 2013) find agg. investment is conditionally heteroskedastic, state dependent response, can be rationalized with lumpy investment. Similar results by Winberry (AER, 2021).
- **Which margin is more important - Extensive vs Intensive?**
    - Guorio and Kashyap (2007) show that it is the number of firms making large investments (extensive margin) that determines bulk of aggregate variation rather than the average size of the spike (intensive margin).

Our Work

What We Do

- Existing literature use General Equilibrium theoretical models.
- In contrast, we use a “novel” empirical approach that combines:
  - (1) A Dynamic Factor Model (DFM) at the firm level, and
  - (2) A Vector Auto-Regression (VAR) at the aggregate level

- Identify business cycle shocks and map their reponses to aggregated/disaggregated firm outcomes.

- Our empirical evidence shows that:
  - **Lumpy adjustments significantly impact aggregate outcomes**
  - **Extensive margin dominates**

Data

Data

- U.S. firm (public) level data:
    - Compustat. Around 14,400 firms covering years 1980-2022.
    - Annual frequency
- Additionally, we use macro variables like GDP in the VAR.

Econometric Model

Factor Model Overview

- We implement a DFM-FAVAR approach to link firm-level dynamics with aggregate outcomes.

- Each firm-level variable is decomposed into:
  - A: A common, aggregate component
  - ϵ: An orthogonal idiosyncratic component.

- The baseline decomposition is given by:

$$x_{i,t} = \Lambda_{ij,t} A_t + \epsilon_{i,t}, \quad \epsilon_{i,t} \sim N(0, V_t) \tag{8}$$

Dynamic Factor Model Details

- **Time-Varying Loadings:**
  - The factor loadings evolve over time, following a random walk:

$$\Lambda_{ij,t} = \Lambda_{ij,t-1} + \eta_t, \quad \eta_t \sim N(0, W_t) \tag{9}$$

- **Common Components Evolution:**
  - The vector of common factors evolves according to a VAR process:

$$A_t = \beta A_{t-1} + u_t, \quad u_t \sim N(0, Q) \tag{10}$$

- Equations (1) and (2) jointly define the DFM-FAVAR system with time-varying loadings.

Identification: Lumpy Adjustments

- **Defining Lumpy Adjustments:**
  - Spikes in capital or labor adjustments are classified as lumpy.
  - A capital adjustment is considered lumpy if the investment-to-capital ratio exceeds a threshold:
$$\frac{I_t}{K_{t-1}} > 0.2.$$
  - A similar threshold is applied for labor adjustments.
  - Threshold value consistent with Cooper and Haltiwanger (2006), Guorio and Kashyap (2007) and Görtz et al. (2023).
- **DFM Restrictions:**
  - In practice, we impose either one-to-one restrictions ($\Lambda_{ij,t} = 1$) or exclusion restrictions ($\Lambda_{ij,t} = 0$) on the sequence of firm-level loadings.
  - Additional restrictions address factor rotational and scale indeterminacies.
  - *No restrictions needed when we use observed factors (firm level aggregates).*

Identification: Aggregate Shocks

- We identify two aggregate shocks using a **Max-FEV** share approach

- **Business Cycle Shock**: the structural innovation that maximizes the FEV of a chosen real activity indicator over a finite horizon.

  Real Activity Target variables: (i) Aggregate Firm Sales, (ii) GDP.

- **Financial Shock**: the structural innovation that maximizes the FEV of a financial conditions indicator over a finite horizon.

  Financial Conditions Target variable: Excess Bond Premium

Estimation: State-Space and VAR Implementation

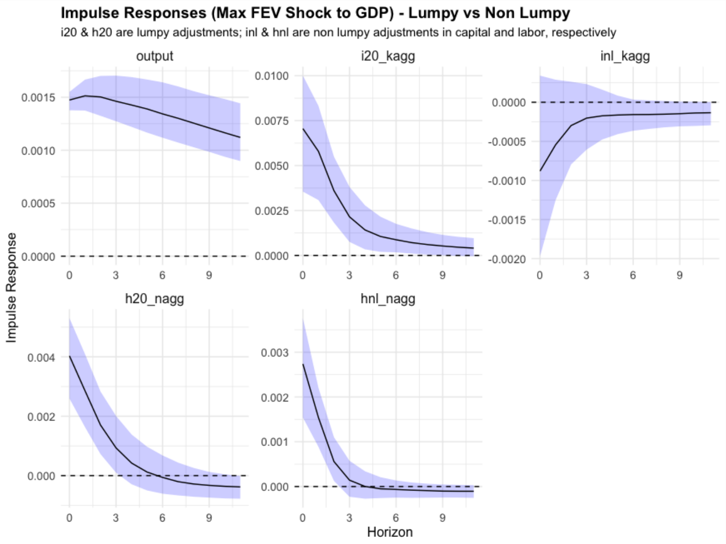
- Equations (1) and (2) define a linear Gaussian state-space model. We obtain Kalman smoothed estimates of the time-varying loadings, $\Lambda_{ij,t}$.
- Variance terms are estimated using an EWMA approach, following Koop and Korobilis (2014).
- The VAR in Equation (3) is estimated using standard VAR procedures.
- Conditional on known (or initialized) values of $\tilde{C}_t$ and estimated parameters $\lambda_{ij,t}$, β, V_t, W_t, and Q, we update the Kalman smoothed estimates for C_t.

Results - Aggregate Dynamics

Impulse Responses - Lumpy vs Non Lumpy - Shock to Sales

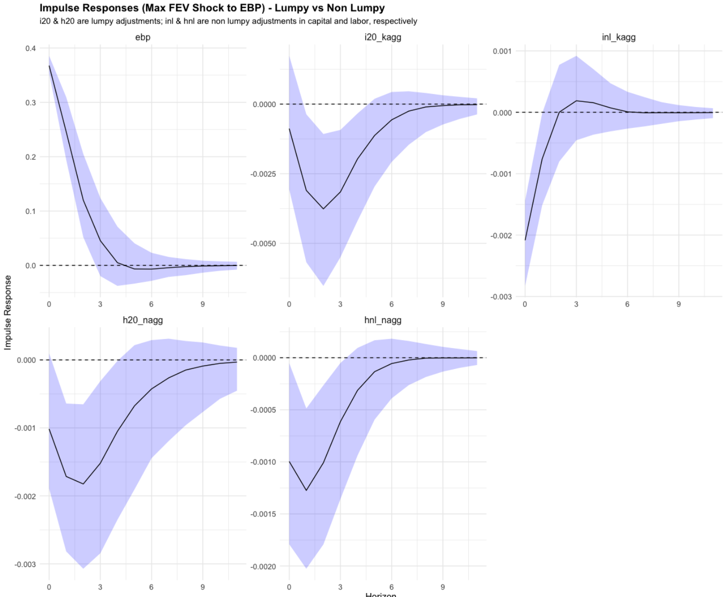
- Response in Lumpy (20) vs Non Lumpy Investment Rate (First Row), and
- Response in Lumpy (20) vs Non Lumpy Employment/Hiring Rate (Second Row)

Impulse Responses - Lumpy vs Non Lumpy - Shock to GDP

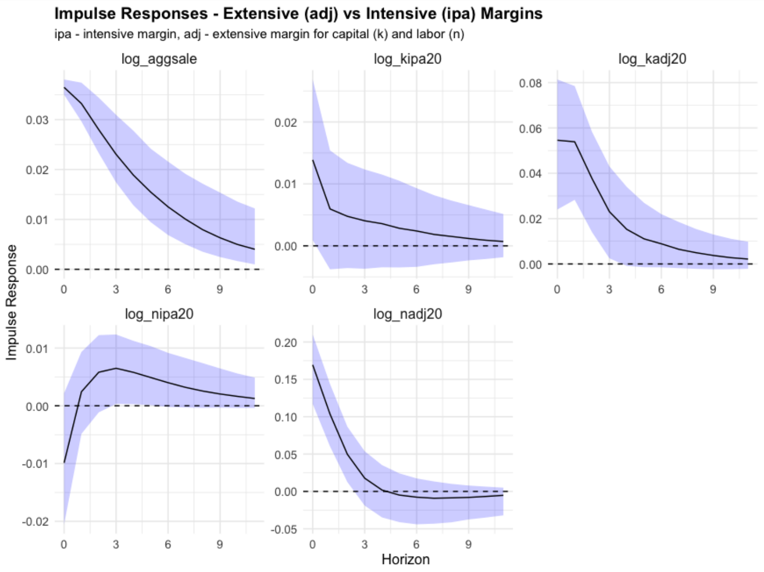


- Significant responses in both lumpy margins.
- Similar Responses when targeting GDP or (Aggregated) Firm Sales.

Impulse Responses - Lumpy vs Non Lumpy - Shock to Excess Bond Premium

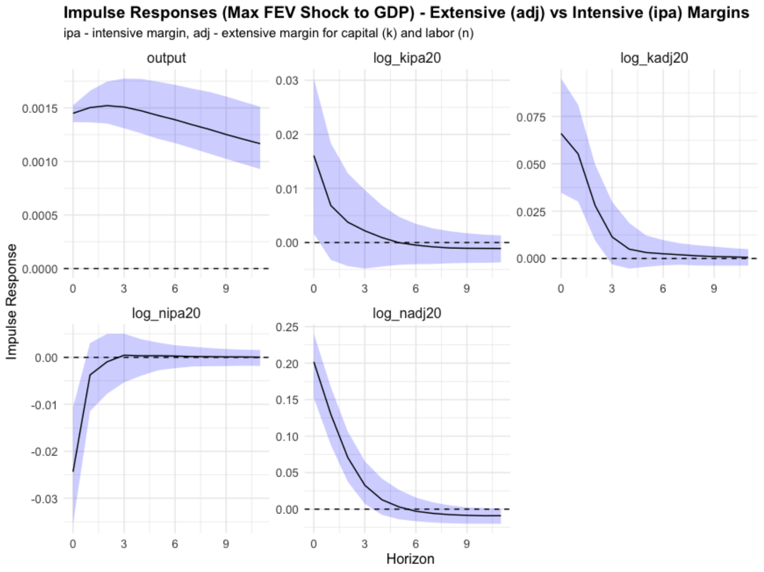


Impulse Responses - Intensive vs Extensive - Shock to Sales



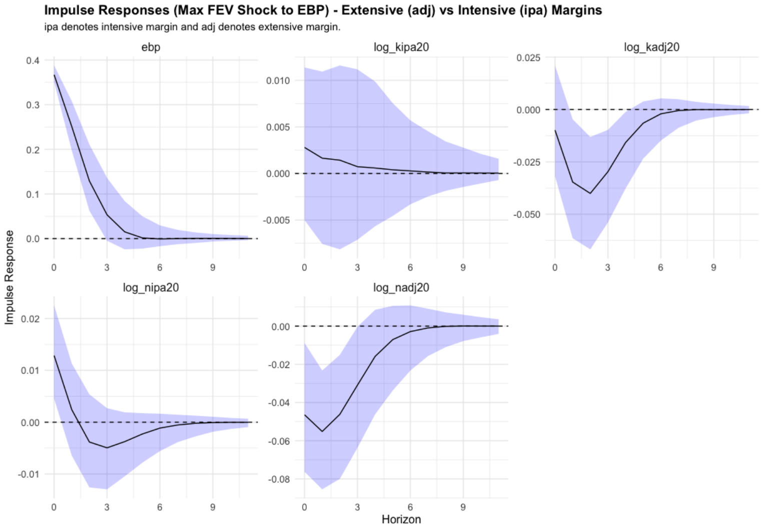
- Intensive (ipa) vs Extensive (adj) adjustments in capital (First Row) and Employment (Second Row)

Impulse Responses - Intensive vs Extensive - Shock to GDP



- Extensive margin is more important.

Impulse Responses - Intensive vs Extensive - Shock to Excess Bond Premium



- Similar pattern emerges: extensive margin drives the adjustment

Results - Firm Dynamics

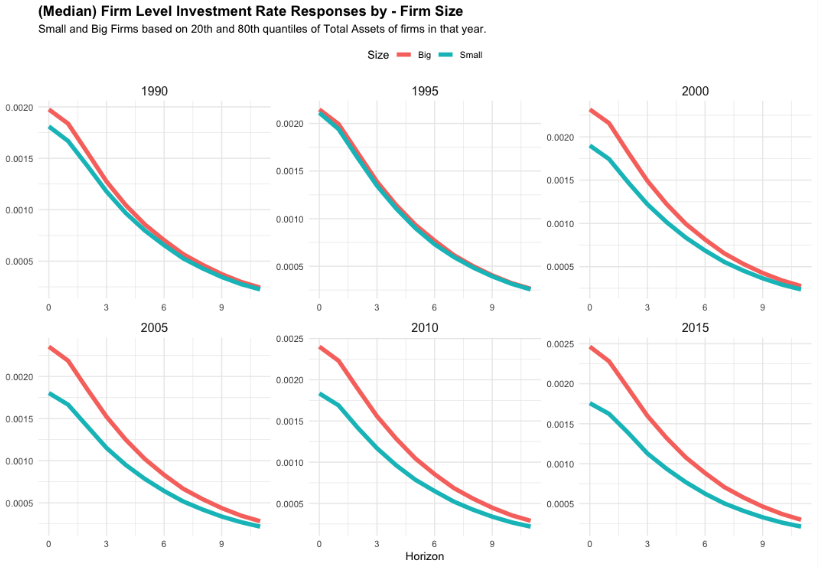
Firm Level Impulse Responses

- Two-way mapping: aggregate response $<==>$ firm level responses.
- Time varying loadings imply time varying impulse responses at the firm level.
- Can group responses impulse responses by firm characteristics:
  - Size:Small vs Big Size Firms.
  - Debt Growth: Low vs High Debt Growth Firms
  - Sales Growth: Low vs High Sales Growth Firms
  - Intangible Intensity: Low vs High Intangible Investment Firms
  - Capital Age: Newer vs Older Capital Vintage Firms - Age
- Type of Aggregate Shock - Shock to EBP

Firm-Level Responses: Summary

- We map firm-level investment responses to aggregate shocks using the DFM-implied investment rate ($x_{i,t}$). This allows a direct micro–macro linkage between aggregate impulses and heterogeneous firm behavior.
- Responses vary systematically across the firm distribution. Larger firms, high-growth firms, and firms with younger capital vintages exhibit stronger and more pronounced investment-rate responses.
- Financial shocks (EBP shocks) reveal substantial cross-sectional dispersion. Small (and financially constrained firms) display negative or muted investment responses, while large firms show minimal or no contraction, highlighting the role of financial frictions.
- Heterogeneity becomes even more pronounced when examining extreme (quantile-based) firm groupings:
  - **Size:** Very Small vs. Very Large Firms
  - **Sales Growth:** Very Low vs. Very High Sales-Growth Firms

(Median) Investment Rate Response at the Firm Level - By Size (I)



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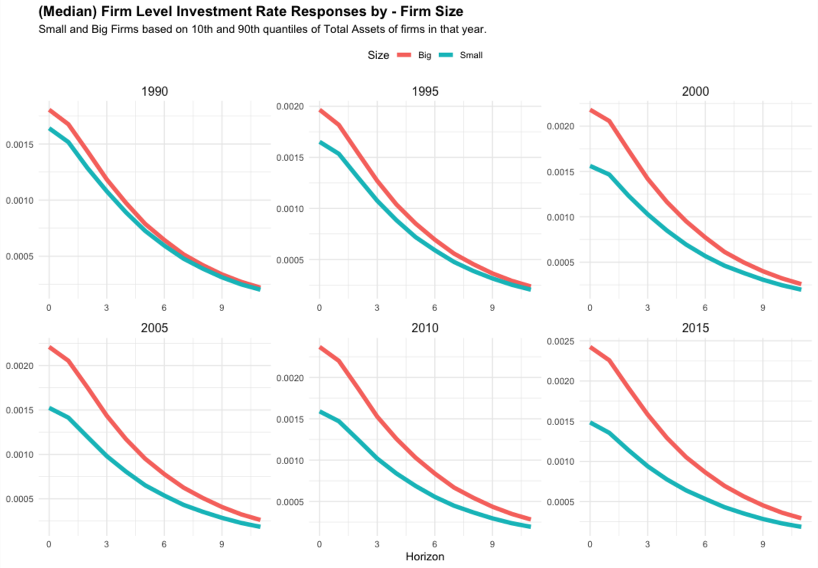
(Median) Investment Rate Response at the Firm Level - By Size (II)



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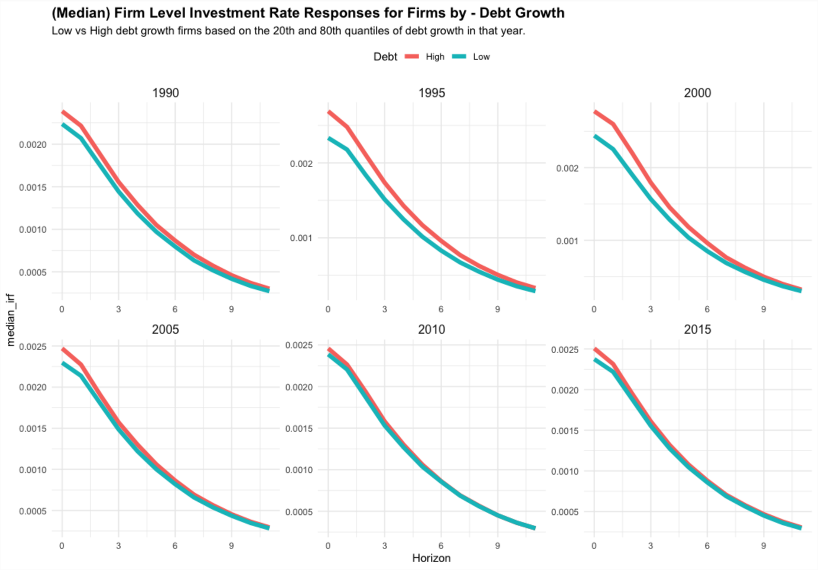
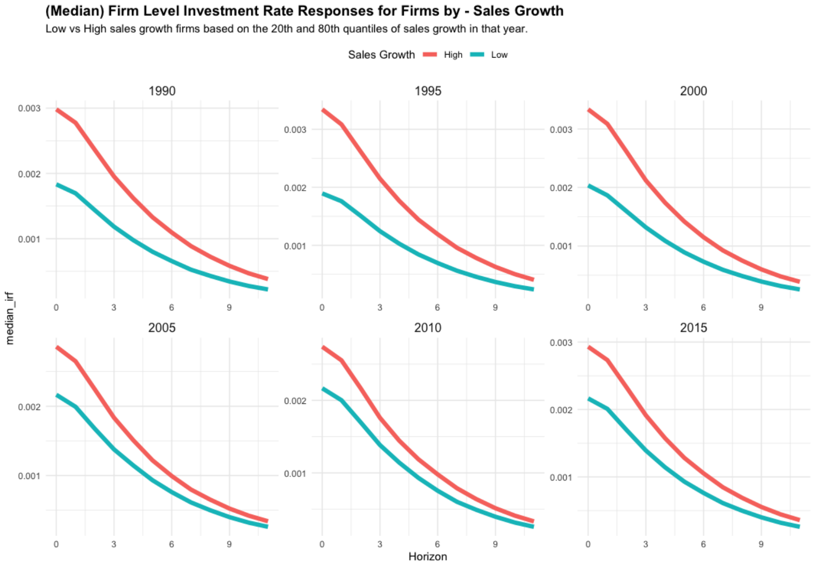
Investment Rate Response at the Firm Level - By Debt Growth



More pronounced effects as you go to more extreme quantiles. High debt

Investment Rate Response at the Firm Level - By Sales Growth (I)



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Investment Rate Response at the Firm Level - By Sales Growth (II)

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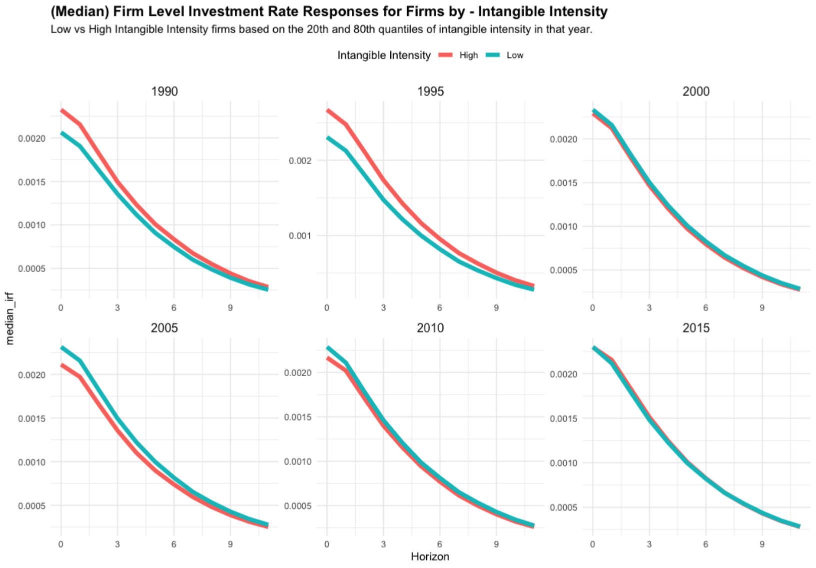
Investment Rate Response at the Firm Level - By Intangible Intensity



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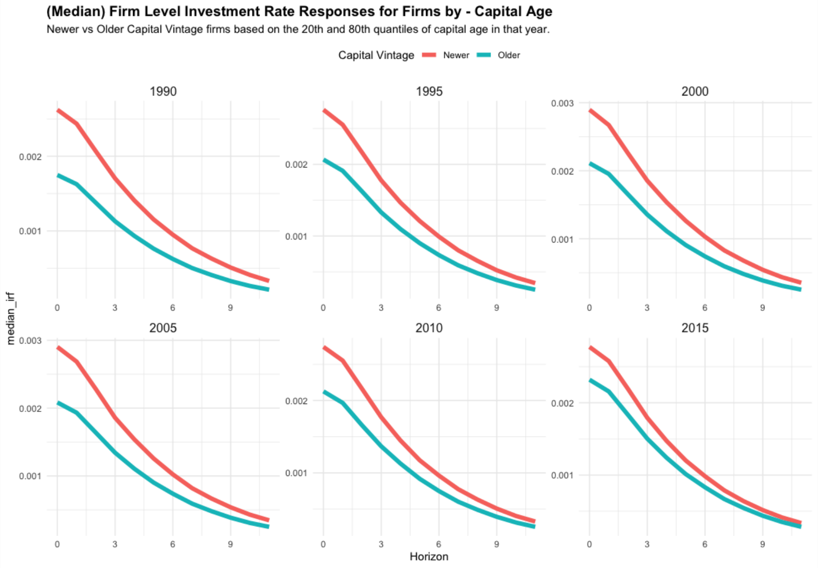
Investment Rate Response at the Firm Level - By Installed Capital Age



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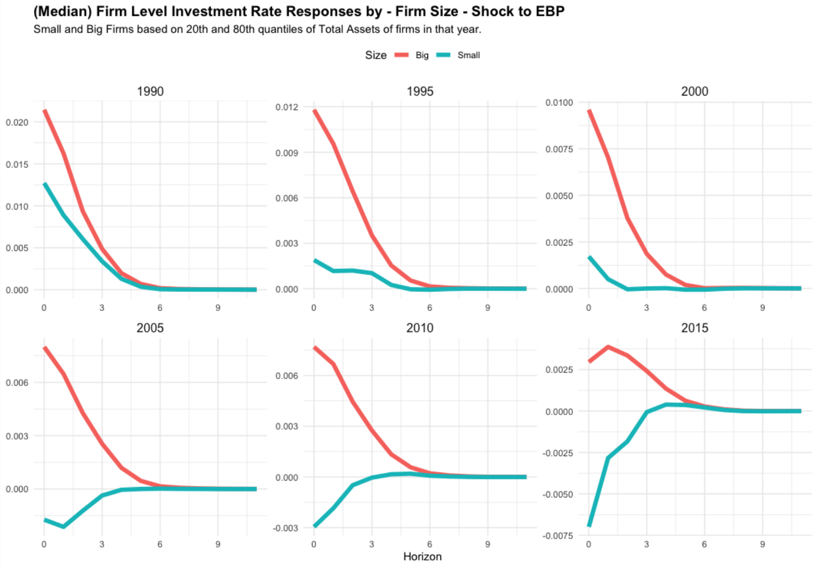
Investment Rate Response at the Firm Level - By Firm Size - Aggregate EBP Shock



Results from Khan Thomas (2008) Simulated Data

Khan–Thomas (2008) Model: Core Idea

- Heterogeneous plants with predetermined k, aggregate shock A, idiosyncratic productivity z, and a **stochastic fixed adjustment cost** ξ drawn each period (the key nonconvexity). Small "maintenance" investments avoid the fixed cost.
- Each plant compares the value of **adjusting to its target** k^* (paying ξ) vs. **not adjusting** (letting $(1-\delta)k$ prevail). This yields a threshold rule: adjust iff $\xi \leq \xi^*(z, k)$.
- Lumpiness arises because depreciation pulls inactive plants away from their targets; occasional low-ξ draws trigger **large jumps** in capital. Idiosyncratic shocks spread adjustment timing across plants.
- Main contribution: a DSGE model consistent with micro evidence—spikes, asymmetry, limited inaction—while showing that in general equilibrium the aggregate impact of lumpiness is **small** due to price-induced smoothing.

Equilibrium Conditions

- Firms choose labor and capital (adjust or not) to maximize their value given p and the shock A.
- Representative household supplies labor inelastically and owns all firms; p is the marginal value of output in units of consumption.

Aggregate Firm Value

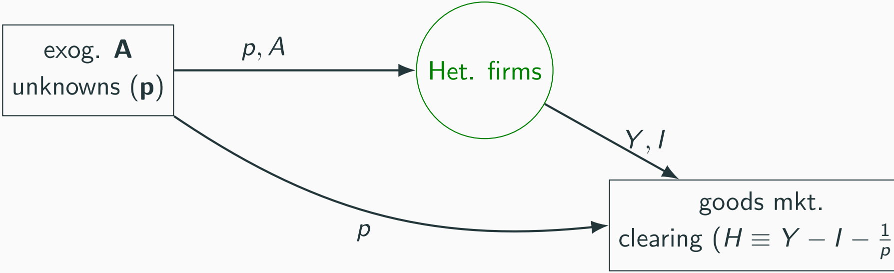
$$V(z,k) = -\phi \int_0^{\xi^*(z,k)} \xi \, dG(\xi) + G(\xi^*) V^A(z,k) + \left[1 - G(\xi^*)\right] V^{NA}(z,k)$$

Goods-Market Clearing (GE Condition)

$$\frac{1}{p} = \iint A\, zk^{\alpha} n(z,k,\xi)^{\nu} \, d\mu(z,k)\, dG(\xi) - \iint \left(k'(z,k,\xi) - (1-\delta)k\right) d\mu(z,k)\, dG(\xi)$$

- $\mu(z,k)$ is the stationary distribution over firm states.
- **Equilibrium**: price p solves the system $H(p,A) = 0$ given firm policies and μ.

Model in DAG Form



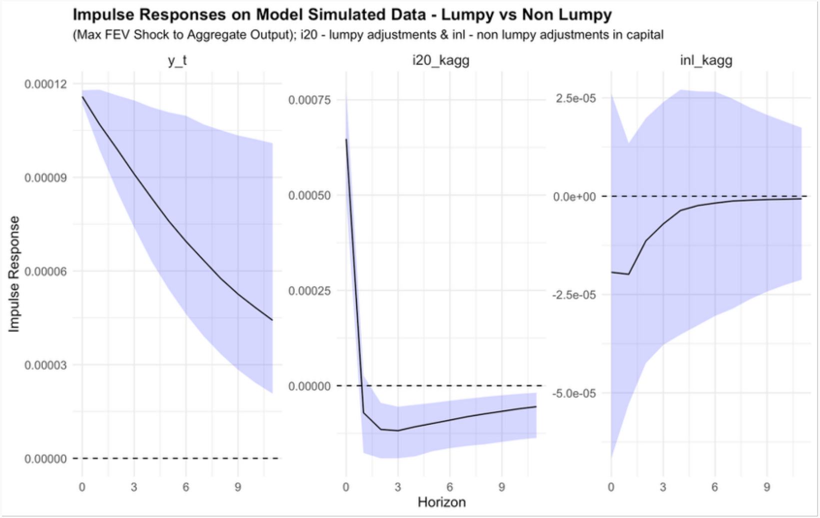
The market equilibrium condition is

$$H(p, A) = 0$$

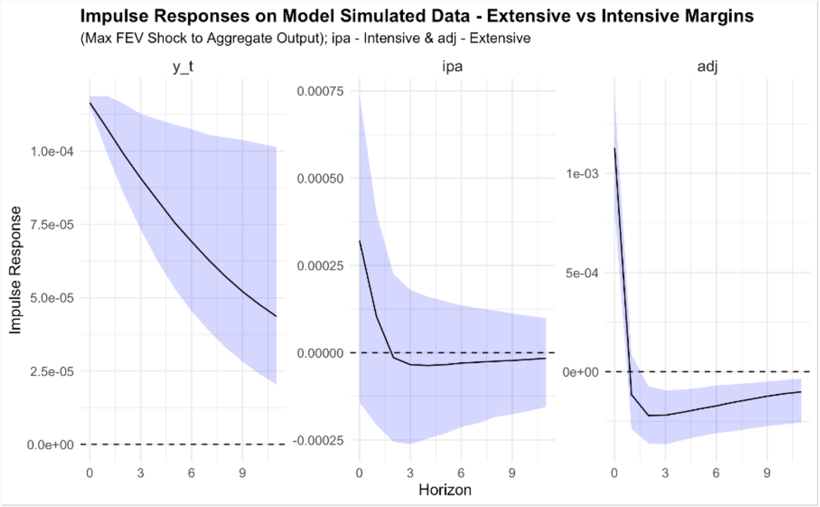
where $H(p, A) = Y - I - \frac{1}{p}$. This sequence formulation enables the solution of the model quickly using the Sequence Space Jacobian algorithm of Auclert et al (2021).

After solving the model, we simulate panel data for 8000 firms and a lot of years. DFM-FAVAR model is then estimated on this data (T=50 years) and impulse responses to aggregates are plotted

Simulated Data - Lumpy vs Nonlumpy Responses



Simulated Data - Extensive vs Intensive Margin Responses



Conclusion

- Using rich firm-level data, we find strong evidence that **aggregate business cycle movements are driven by lumpy adjustment dynamics at the firm level**.
- The **extensive margin**—the share of firms undertaking large capital changes—**plays a far more important role than the intensive margin** (the size of such changes).
- Both business cycle and financial shocks generate **heterogeneous firm-level responses** that vary systematically by size, leverage, and investment intensity.
- Lumpy adjustments in capital and labor frequently occur **synchronously** within firms, amplifying aggregate comovement.
- The findings carry important policy implications related to monetary policy, business dynamism and growth.